

Summer Science Institute 2025



AUBURN
UNIVERSITY

College of Sciences
and Mathematics

Description: This residential summer science program for outstanding 11th-12th grade students interested in science and mathematics is open to students residing in Alabama or Georgia. Seating was limited to 16 students and participation granted on an academically competitive basis. During the program students engaged with real-world applications and practitioners of science, performed experiments using cutting edge research equipment, and partnered with COSAM researchers to gain lab skills not taught in high school.

Dates: Sunday, June 1, 2025 – Saturday, June 7, 2025

Personnel:

Faculty Coordinator: Dr. Allen Landers (Physics)

Faculty Fellow: Dr. Rachel Prado (Chemistry and Biochemistry)

Counselors: Megan Fitzgerald (PhD Student: Biology), William Hudson (PhD Student: Chemistry and Biochemistry), Avery Klose (Rising Senior: Biomedical Sciences and Secondary Science Education, COSAM Outreach AUTEACH Intern), and Mason Sake (PhD Candidate: Physics).

COSAM Faculty:

Department of Biological Sciences: Dr. Kate Buckley, Dr. Rita Graze, Dr. Geoff Hill, Shawn Jacobsen, Dr. Aaron Rashotte

Department Chemistry and Biochemistry: Dr. Chris Greico, Dr. Ethan Hill, Dr. Evangelos Miliordos, Dr. Paul Ohno, Dr. Filip Pawlowski, Dr. Wei Zhan

Department of Physics: Dr. Steve Bromley, Dr. Micheal Gramlich, Dr. Hong Zhao

Department of Mathematics and Statistics: Dr. Le Chen

Department of Geosciences: Dr. Stephanie Shepherd

Other AU Faculty and COSAM Outreach Office Staff: Mary Lou Ewald (Director), Dr. Jessica Gilpin (Assistant Director)





Impact:

- **Total Number of Students:** 16 students (8 male, 8 female)
- **Age Range:** rising 11th – 12th grades

Counties Impacted:

- **Alabama:** Madison, Montgomery, Coffee, Lee, Shelby, Etowah, Tallapoosa
- **Georgia:** Gwinnett, Fulton



Auburn Facilities: Academic Classrooms and Laboratories Complex, Auburn University Aviary, Donald E. Davis Arboretum, Haley Center, Hay Day Market, Leach Science Center, Melton Student Center, Boyd Residence Hall, Sciences Center Auditorium, Science Center Classrooms, Science Center Laboratories

The 2025 Summer Science Institute hosted 16 highly motivated, high achieving students who were chosen to participate by a competitive application process. The average ACT score of the participants was **32.7** with a range of 28 to 36 (PSAT average – **1406.2**, range – 1200-1500). Prior to the first evening of the program, the students completed a short, informational pre-program Qualtrics survey. On the last day of the program, all 16 participants responded to a post-program survey. The following section highlights some of the key results from the surveys.

Survey Results:

- **100%** of students reported that SSI positively impacted their summer.
- **100%** of students reported that they were either somewhat or very satisfied with the overall experience, the interactions with and helpfulness of counselors, and the interactions with faculty.
- **88.9%** of students who have attended other summer academic programs indicated that SSI was more enjoyable than other academic summer programs and **66.7%** indicated that they learned more content during SSI than those other programs.
- **87.5%** of students reported that participation in SSI has helped them better determine their future career choice.
- **75%** reported their interest in attending Auburn University has increased.
- After participating in SSI, **75%** reported an increased interest in biology, **56%** reported an increased interest in chemistry, **56%** reported an increased interest in physics, **38%** reported an increased interest in mathematics, and **31.2%** reported an increased interest in geology.



- After participating in SSI, **88%** of students reported an increased awareness and knowledge of careers in biology, **94%** in chemistry, **75%** in physics, **88%** in geosciences, and **69%** in mathematics.
- After participating in SSI, **94%** of students reported an increased awareness of current research topics in biology, **88%** in chemistry, **88%** in geosciences, and **94%** in physics, and **63%** in mathematics.
- **100%** reported that participation in SSI has increased or greatly increased their understanding of how to engage in scientific research.

- **94%** indicated that their interest in becoming a scientist has increased, which is a **31% increase** from the previous year.
- The most enjoyable afternoon/evening activities were the Chemistry Demo Night and the Games and Band Night at VCOM, Both with a rating of **4.75** out of **5**. The next most enjoyed activity was the Grill Out with games and a movie at VCOM with a score of **4.69** out of **5**.

Module Report:

- Students reported the most enjoyable academic session was the Chem Demo Show with Dr. Hill, with an average rating of **4.75** out of **5**. The next most enjoyable sessions were the Herp Hunt with Shawn Jacobsen and “Changing Color with Electricity” with Dr. Grieco, with average ratings of **4.63** and **4.5** out of **5**, respectively.

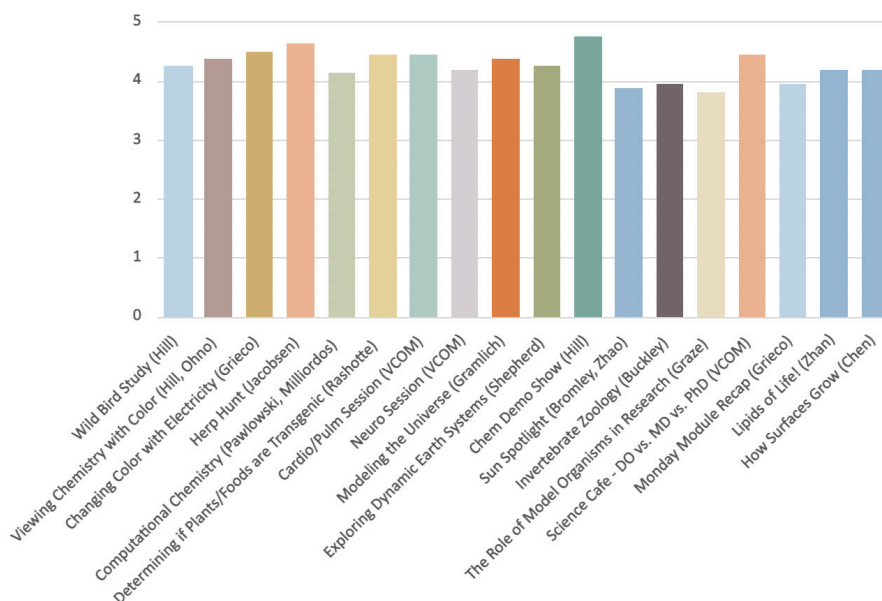
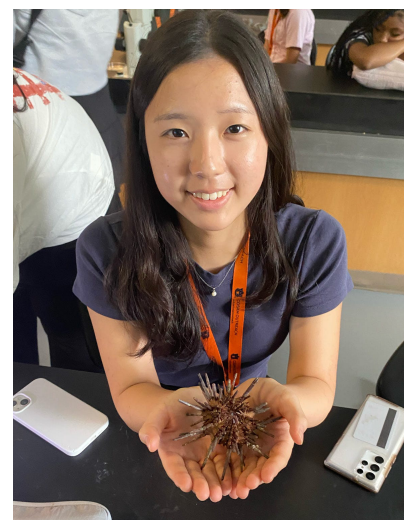


Figure 1. Distribution of how enjoyable the students felt each academic session was while participating in SSI.



- Students felt they acquired the most content knowledge in the Cardio/Pulmonary Session at VCOM with an average rating of **4.375** out of **5**, followed by “Modeling the Universe” with Dr. Mike Gramlich with an average rating of **4.25** out of **5**.

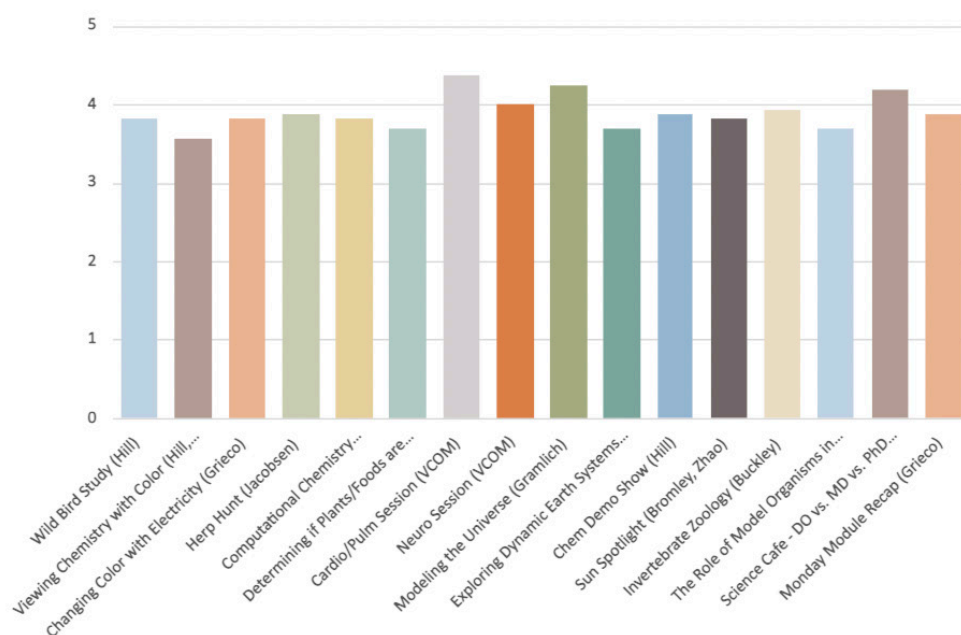


Figure 2. Distribution of content knowledge acquired during SSI academic modules.

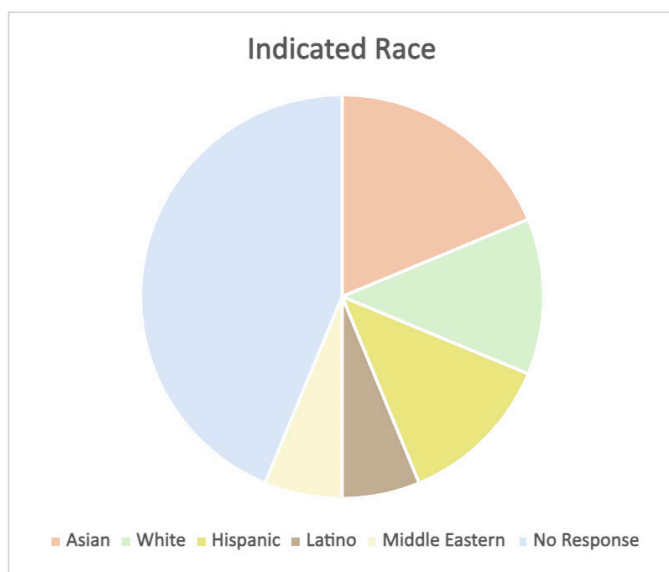




Participant Demographics:

Out of 16 participants, there were eight male students and eight female students, with two rising juniors and fourteen rising seniors. Two participants attend a Title I high school.

The distribution of their indicated race is illustrated below.



Financial Support:

All expenses are covered for SSI students thanks to support from the College of Sciences and Mathematics Outreach Center and the National Science Foundation.

SSI Schedule 2025

Week of: June 1

	6/1 SUNDAY	6/2 MONDAY	6/3 TUESDAY	6/4 WEDNESDAY	6/5 THURSDAY	6/6 FRIDAY	6/7 SATURDAY	
7:00 AM		Breakfast	Breakfast	Breakfast	Breakfast	Breakfast	Dorm Checkout with Parents	
7:30 AM								
8:00 AM								
8:30 AM		Geoff Hill (8:30-11:30) AU Aviary	Evangelos Miliordos Filip Pawlowski (8:30-11:30) ACLC 153	VCOM with SEE (8:00-12:00) VCOM Anatomy Lab 420	Hong Zhao Steve Bromley (8:30-11:30) Leach 3142	CWE COSAM Browse (8:45-9:15) ACLC		
9:00 AM						Chris Grieco (9:30-10:30) Chem 251		
9:30 AM								
10:00 AM								
10:30 AM					Wei Zhan (10:40-11:40) SCL 231			
11:00 AM								
11:30 AM								
12:00 PM		LUNCH	LUNCH	LUNCH	LUNCH	Le Chen (11:50-12:50) SCA		
12:30 PM								
1:00 PM		Paul Ohno	Aaron Rashotte	Mike Gramlich		Lunch		
1:30 PM		Ethan Hill	(1:00-3:00)	(1:15-3:15)	Kate Buckley and Rita Graze w/ VCOM	Faculty Meet and Greet		
2:00 PM		(1:00-3:00)	SCL 231	Leach 1109	(1:15-3:15)	(1:00-2:30) ACLC 259 A		
2:30 PM		SCL 105			SCL 231 and 227			
3:00 PM	Check-In (Boyd Hall)	Chris Grieco		Steph Shepherd		Bookstore Trip (2:45-3:30)		
3:30 PM		(3:00-5:00)	VCOM with SEE (3:30-5:00)		(3:30-5:30)	Kate Buckley Rita Graze	Final SSI Jeopardy (3:30-4:30) ACLC 153	
4:00 PM		SCL 105	VCOM Anatomy Lab 420		ACLC 226	(3:30-5:30)		
4:30 PM						SCL 231 and 227	Dinner (Beyond the Wok)	
5:00 PM								
5:30 PM		Dinner (SCA)	Dinner Grill out at VCOM VCOM - Patio		Dinner (VCOM) MPR	Summer Nights in Downtown Auburn (6-10pm)		
6:00 PM	Welcome Dinner Get To Know You Games Tour of VCOM (VCOM - MPR 126)	Down Time at Dorm		Taco Night (6pm) at Arboretum then Chemistry Demo Show outside of the Chem Building starting ~7:30pm	Science Cafe at VCOM (DO vs. MD vs. PhD, lightening talks)			
6:30 PM								
7:00 PM								
7:30 PM								
8:00 PM		Herp Hunt with Shawn Jacobsen in Tuskegee National Forest (Meet at Dorm)	Movies and Yard and Board Games VCOM - MPR 126 and patio/lawn		Music and Yard Games VCOM - MPR 126 and patio/lawn			
8:30 PM								
9:00 PM								
9:30 PM								
10:00 PM								
10:30 PM								
11:00 PM	Lights Out	Lights Out	Lights Out	Lights Out	Lights Out	Lights Out		

Dorm Counselors:

1) Avery

2) Megan

3) Alex

4) Josh

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1) Avery

2) Megan

3) Josh

4) Mason

Dorm Counselors:

1) Megan

2) Alex

3) Josh

4) Mason

Dorm Counselors:

1) Avery

2) Megan

3) Mason

4) Alex

Dorm Counselors:

1) Avery

2) Alex

3) Mason

4) Josh

Dorm Counselors:

1) Avery

2) Megan

3) Alex

4) Josh

Summer Science Institute

Class of 2025

